

🕒 LeetCode Scoring — Deep Dive (40 pts total)

What data we fetch from LeetCode

LeetCode has a **public GraphQL API** (leetcode.com/graphql). The scraper sends two queries:

Query	Fields fetched	What it means
<code>userPublicProfile</code>	<code>totalSolved</code> , <code>easySolved</code> , <code>medSolved</code> , <code>hardSolved</code>	Practice/problem history
<code>userContestRanking</code>	<code>rating</code> , <code>attendedContestsCount</code> , <code>globalRanking</code>	Contest performance

If `userContestRanking` returns `null` → student has **never entered a contest**.

Part 1 — LC Problem Score (20 pts)

This measures **how deeply you've practiced** on LeetCode, weighted by difficulty.

Step 1: Weighted Raw Points

Different difficulties are worth different amounts because **Hard problems signal real skill**:

```
raw_pts = (easy × 2) + (medium × 4) + (hard × 11)
```

Why these weights specifically?

Difficulty	LC difficulty range	Real-world signal	Weight
Easy	Basic DSA, brute force	Can code at all	2
Medium	Arrays, DP, graphs	Placement-ready	4
Hard	Advanced DP, segment trees	FAANG-level	11

Hard is weighted **11** (not **8** or **10**) because the **ratio of skill** between Medium and Hard is non-linear — a student who solves 10 Hard problems is much stronger than one with 20 Medium problems.

Examples:

Student profile	Raw pts
200E + 50M + 0H	200×2 + 50×4 + 0×11 = 600
50E + 100M + 10H	50×2 + 100×4 + 10×11 = 610
10E + 20M + 30H	10×2 + 20×4 + 30×11 = 430

Student profile	Raw pts
500E + 0M + 0H	500×2 = 1000 (Easy grinders)

Step 2: Consistency Factor

We don't have week-by-week submission data from LC's public API. So we **estimate consistency** from total volume — a student who solved 500 problems didn't do it all in one day:

Total solved	Factor	Reasoning
≥ 300	0.85	Very consistent, daily solver
≥ 150	0.75	Regular, solid effort
≥ 50	0.65	Sporadic but trying
< 50	0.55	Just started or inactive

```
effective = raw_pts × consistency_factor
```

Why not 1.0 for ≥ 300? Even the most consistent student gets a 0.85 cap because we can't verify *how* they solved them — contest cheating, copy-paste solutions, etc. are all possible. The cap represents **realistic uncertainty**.

Step 3: Ratio against Benchmark

The benchmark = **450 UDG points** — what a "perfect" LC problem student should accumulate:

```
ratio = clamp(effective / 450, 0, 1)
```

What is 450? It's calibrated for a student who solved roughly:

- 200 Medium problems + 15 Hard problems + some Easy → effective ≈ 450

This is a realistic target for a placement-ready student with ~1.5 years of practice.

Step 4: Power Curve (the most important step)

```
prob_score = 20 × ratio^0.7
```

The **0.7** exponent is **deliberately less than 1** (concave curve). Here's why this matters:

Ratio	Linear (ratio × 20)	Power curve (ratio^0.7 × 20)	Difference
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Ratio	Linear (ratio × 20)	Power curve (ratio^0.7 × 20)	Difference
0.10	2.0	3.5	+1.5
0.20	4.0	6.1	+2.1
0.30	6.0	8.2	+2.2
0.50	10.0	12.1	+2.1
0.75	15.0	16.6	+1.6
1.00	20.0	20.0	0

Effect: Students who've done 30–50% of the benchmark aren't crushed to near-zero. The curve is **fair to mid-range students** while still rewarding top performers maximally.

Full Problem Score Example

Student A — 120E + 80M + 5H solved, total = 205 problems:

```
raw_pts      = (120×2) + (80×4) + (5×11) = 240 + 320 + 55 = 615
factor       = 0.75  (≥150 total)
effective    = 615 × 0.75 = 461.25
ratio        = min(461.25/450, 1.0) = 1.0  (just exceeded benchmark!)
prob_score   = 20 × 1.0^0.7 = 20.0 pts
```

Student B — 40E + 15M + 0H solved, total = 55 problems:

```
raw_pts      = (40×2) + (15×4) + (0×11) = 80 + 60 = 140
factor       = 0.65  (≥50 total)
effective    = 140 × 0.65 = 91
ratio        = 91 / 450 = 0.202
prob_score   = 20 × 0.202^0.7 = 20 × 0.304 = 6.1 pts
```

Student C — 0 problems solved:

```
prob_score = 0
```

Part 2 — LC Contest Score (20 pts)

This measures **how actively you compete** and **how good you are at timed contests**.

```
contest_score = 20 × (0.40 × P + 0.60 × L)
```

Two sub-components with different importance:

Component	Weight	What it measures
P (Participation)	40%	Did you actually show up?
L (Level/Rating)	60%	How good are you when you show up?

Why L > P? Because entering 40 contests at rating 1200 is less impressive than entering 5 contests and reaching rating 1900. Quality > quantity for contests.

P — Participation

```
P = clamp(contest_count / 20, 0, 1)
```

Benchmark = **20 contests** (roughly 1 per month over 1.5 years — LC holds weekly contests).

Contests attended	P value	Full participation points (0.40 × P × 20)
0	0.00	0.0
5	0.25	2.0
10	0.50	4.0
15	0.75	6.0
20+	1.00	8.0

L — Level from Rating (the critical one)

The rating is mapped to a **0–1 level score** using linear interpolation between anchors:

LC Rating	L value	What it means
≤ 1400	0.05	Participated but never placed
1500	0.20	Average performer
1600	0.35	Consistent top-half finisher
1700	0.50	~Top 25% consistently
1800	0.65	Very strong, solves 3/4 problems
1900	0.80	Near-expert, sometimes solves all 4
2100+	1.00	Top 1%, expert/master level

Linear interpolation between anchors — e.g., rating 1650:

```

L = 0.35 + (1650 - 1600) / (1700 - 1600) × (0.50 - 0.35)
  = 0.35 + 0.5 × 0.15
  = 0.35 + 0.075
  = 0.425

```

IMPORTANT — BUG-10 fix applied:

```

if contest_count == 0 → L = 0 (not 0.05)

```

Before the fix, a student who **never entered a single contest** still got $L = 0.05$ (the floor anchor). That was wrong — no participation = no level score at all.

Full Contest Score Example

Student: rating = 1714, contests = 24

```

P = min(24/20, 1.0) = 1.0
L = lerp(1700→0.50, 1800→0.65, at rating=1714)
  = 0.50 + (14/100) × (0.65 - 0.50)
  = 0.50 + 0.021
  = 0.521

contest_score = 20 × (0.40 × 1.0 + 0.60 × 0.521)
               = 20 × (0.40 + 0.313)
               = 20 × 0.713
               = 14.26 pts

```

Student: rating = 1450, contests = 8

```

P = 8/20 = 0.40
L = lerp(1400→0.05, 1500→0.20, at rating=1450)
  = 0.05 + (50/100) × (0.20 - 0.05)
  = 0.05 + 0.075
  = 0.125

contest_score = 20 × (0.40 × 0.40 + 0.60 × 0.125)
               = 20 × (0.16 + 0.075)
               = 20 × 0.235
               = 4.7 pts

```

Student: 0 contests ever

```
P = 0, L = 0
contest_score = 20 × 0 = 0 pts
```

How to maximize your LC score

Action	Points gained
Solve your first 50 problems (any difficulty)	~6–8 problem pts
Get from 50 → 150 total (Medium focused)	+4–6 more pts
Enter your first 5 LC contests	+2.0 contest pts (just for showing up)
Get rating from 1400 → 1600	L goes 0.05 → 0.35, +3.6 pts
Get rating from 1600 → 1800	L goes 0.35 → 0.65, +3.6 pts
Get rating from 1800 → 2000	L goes 0.65 → 0.95, +3.6 pts
Solve 10 Hard problems (instead of 30 Easy)	+75 raw pts (same effort, 3.75× reward)

Biggest bang for effort: Enter contests regularly. Even a rating of 1600 with 20 contests = ~11.2 contest pts out of 20. Just showing up beats most students who never contest.